

3. Para $A \in \mathbb{R}^{n \times n}$ simétrica definida positiva, $\|x\|_A = \sqrt{x^T A x}$ y $x^* = A^{-1}b$, sean

$$f_1(x) = \frac{1}{2} \|x - x^*\|_A^2, \quad f_2(x) = \frac{1}{2} \|Ax - b\|_{A^{-1}}^2, \quad f_3(x) = \frac{1}{2} x^T A x - x^T b.$$

Calcule el gradiente de cada función. Concluya que estas funciones difieren en a lo sumo una constante.

Voy a hacer f_2

$$\begin{aligned} f_2(x) &= \frac{1}{2} \|Ax - b\|_{A^{-1}}^2 = \frac{1}{2} (Ax - b)^T A^{-1} (Ax - b) = \\ &= \frac{1}{2} (x^T \underbrace{A A^{-1}}_I - b^T A^{-1}) (Ax - b) \\ &= \frac{1}{2} (x^T A x - \underbrace{b^T A^{-1} A x}_I - x^T b + b^T A^{-1} b) \end{aligned}$$

$$\Rightarrow f_2(x) = \frac{1}{2} x^T A x - b^T x + \frac{1}{2} b^T A^{-1} b$$

Derivamos respecto de x_k , $k=1, 2, \dots, n$:

$$\frac{\partial f_2(x)}{\partial x_k} = \frac{1}{2} \underbrace{\frac{\partial}{\partial x_k} (x^T A x)}_{(1)} - \underbrace{\frac{\partial}{\partial x_k} (b^T x)}_{(2)} + \underbrace{\frac{\partial}{\partial x_k} (b^T A^{-1} b)}_{(3)}$$

$$\textcircled{1} \frac{\partial}{\partial x_k} (x^T A x) = \frac{\partial}{\partial x_k} \left(\sum_{i=1}^n x_i \underbrace{(Ax)_i}_{(*)} \right) = \frac{\partial}{\partial x_k} \left(\sum_{i=1}^n \sum_{j=1}^n x_i a_{ij} x_j \right)$$

def de
prod. Interno



$$\left[\sum_{j=1}^n a_{ij} x_j \right]$$

$$= \sum_{i=1}^n \sum_{j=1}^n \frac{\partial}{\partial x_k} (x_i a_{ij} x_j) \text{ Sigue m } \otimes$$

$$Ax = \begin{bmatrix} \vdots \\ \sum_{j=1}^n a_{kj} x_j \\ \vdots \end{bmatrix}$$

Cuenta auxiliar:

$$\frac{\partial}{\partial x_k} x_i a_{ij} x_j = \begin{cases} 2a_{kk}x_k, & \text{Si } i=j=k \\ a_{kj}x_k, & \text{Si } i=k \neq j \\ a_{ik}x_k, & \text{Si } i \neq k=j \\ 0 & \text{Si } i,j \neq k \end{cases}$$

$$\otimes \quad 2a_{kk}x_k + \sum_{\substack{j=1 \\ j \neq k}}^n \underbrace{a_{kj}}_{\text{A es simétrica}} x_k + \sum_{\substack{i=1 \\ i \neq k}}^n \underbrace{a_{ik}}_{\text{A es simétrica}} x_k = 2 \sum_{j=1}^n a_{kj} x_k$$

Ento vale para todo $k = 1, 2, \dots, n$:

$$\frac{\partial f(x)}{\partial x_k} = 2 \sum_{j=1}^n a_{kj} x_j$$

$$\Rightarrow \nabla(x^T A x) = A x \quad \checkmark$$

$$\textcircled{2} \quad \frac{\partial}{\partial x_k} (b^T x) = b_k \quad \forall k$$

$$\textcircled{3} \quad \frac{\partial}{\partial x_k} (b^T A^{-1} b) = 0 \quad \forall k$$

Luego,

$$\nabla f_2(x) = Ax - b \quad \checkmark$$

